National University of Computer and Emerging Sciences, Lahore Campus

Course:	Complex Variables and Transforms	Course Code:	MT 220
Program:	EE	Semester:	Fall-17
Duration:	60 min.	Total Marks:	25
Paper Date:	18-09-2017	Weight	12.5%
Section:	All sections	Page(s):	4
Exam:	Mid 1	Roll No:	15-4561

Instruction/Notes: Exchange of calculators or programmable calculators are not allowed at all. Attempt both questions on this question paper. Extra sheet may be used for rough work/calculations (Do not attach). Do all steps neat & clear.

1. What is harmonic function? Prove that the function $v = xy$ is harmonic. Also find the corresponding conjugate u and hence the original analytic function $f(z) = u + iv$, where $i = \sqrt{-1}$

(2+2+4+2=10 Marks, CLO-2)

(a) Harmonic functions:

A function $f(z)$ is said to be harmonic function in Domain "D" if the function is analytic in the Domain "D" and

(i) partial derivatives exist in Domain

(ii) satisfy the Laplace's equation

10

(b) $v = xy$

Sol:-

$V_x = y \Rightarrow V_{xx} = 0$ $\qquad \nabla^2 V = V_{xx} + V_{yy} = 0$

$V_y = x \Rightarrow V_{yy} = 0$ $\qquad$ Hence, The function is harmonic.

As we know that

CR-Equations:-

$u_y = -V_x \rightarrow ①$

$u_x = V_y \rightarrow ②$

From Equation ②

$u_x = V_y = x$

$\Rightarrow u_x = x$

$\int u_x \, dx = \int x \, dx$

National University of Computer and Emerging Sciences, Lahore Campus

Course:	Complex variables and Transforms	Course Code:	MT 220
Program:	EE	Semester:	Fall-17
Duration:	3 hrs.	Total Marks:	70
Paper Date:	14-12-2017	Weight	50%
Section:	All sections	Page(s):	1
Exam:	Final	Roll No:	

Instruction/Notes: Answer all questions neatly. No questions and no comments. Each question carries 10 points.

1. Define a harmonic function. Is the function $u = e^{-x}\sin 2y$ harmonic? If your answer is yes, then find the analytic function by finding the corresponding conjugate.

2. A voltage $v(t) = V_0 \cos 100\pi t$, where t is the time, is passed through a half wave rectifier that clips the negative portion of the wave. Develop a clear sketch of the wave and then find the Fourier series of the resulting periodic function.

3. State Cauchy's Residue Theorem. Using the theorem, evaluate the real integral

$$\int_0^{2\pi} \frac{\cos(x)}{\frac{5}{4} - \cos(x)}\, dx$$

4. Find Fourier Transform of the signal represented by the following function

$$f(t) = \begin{cases} e^{-2t} & if \quad t > 0 \\ 0 & otherwise \end{cases}$$

5. Find the impedance of a circuit which is given by the solutions of $\cosh z = -1$.

6. Show that the following integral represents the indicated function

$$\int_0^{\infty} \frac{\sin w - w\cos w}{w^2} \sin xw \, dw = \begin{cases} \frac{1}{2}\pi x & if \quad 0 < x < 1 \\ \frac{1}{4}\pi & if \quad x = 1 \\ 0 & if \quad x > 1 \end{cases}$$

7. For the contour integral,

$$\int_C \sec^2 z \, dz$$

where c is a straight line from $\pi/4$ to $\pi i/4$,

i) Is the integrand analytic? Explain.
ii) Is there any point or points of singularities? If yes, then find.
iii) Solve the integral using any appropriate method.

2. The impedance of a circuit is given by the complex number Z satisfying the equation $e^z = \frac{7i+1}{1+i}$.

   a. Solve the above equation to find the resistance (real part of impedance) and reactance across an inductor (imaginary part of impedance).

   b. Find the polar form of the given complex number.

   c. Using the answers of part (a) or part (b), calculate the ratio of the voltage difference amplitude to the current amplitude (magnitude of the complex number) and the phase difference between voltage and current (the argument of the complex number).

   d. Draw an argand diagram to give a pictorial description of the answers to the above part.

(7+3+3+2=15 Marks, CLO-1, 3)

(a) $e^z = \frac{7i+1}{1+i}$

Sol:-

$$e^z = \frac{7i+1}{1+i}$$

$$e^z = \frac{7i+1}{1+i} \times \frac{1-i}{1-i}$$

$$e^z = \frac{(7i+1)(1-i)}{1-i^2}$$

$$e^z = \frac{8+6i}{2}$$

$$e^z = 4+3i$$

where

$$z = x+iy$$

$$e^z = e^{x+iy}$$

$$e^{x+iy} = 4+3i$$

$$|e^z| = e^x = 5$$

$$e^x = 5$$

$$\boxed{x = \ln 5}$$

$$e^x(\cos y + i\sin y) = 3i+4$$

$$e^x \cos y + i e^x \sin y = 4+3i$$

$$\tan y = \frac{\sin y}{\cos y} = \frac{3}{4}$$

$$y = \tan^{-1}\frac{3}{4}$$

$$z = x+iy$$

$$\boxed{z = \ln 5 + i\tan^{-1} 3/4}$$

& $Re(z) = \ln 5$

$Im(z) = \tan^{-1} 3/4$

15

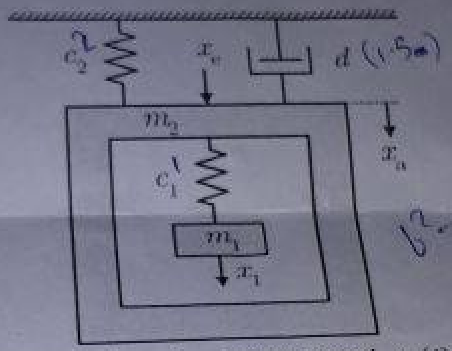
National University of Computer and Emerging Sciences, Lahore Campus

	Course:	Complex variables and Transforms	Course Code:	MT 226
	Program:	EE	Semester:	Fall-17
	Duration:	60 min.	Total Marks:	30
	Paper Date:	02-11-2017	Weight	15%
	Section:	All sections	Page(s):	
	Exam:	Mid 2	Roll No:	

Instruction/Notes: Answer all questions neatly with clear figures where required. Exchange of calculators or programmable calculators are not allowed at all.

1. The mass-spring-damper mechanical system in Figure with the mechanical constants $c_1 = 1$, $c_2 = 2$, $d = 1.5$, $m_1 = 1$ and $m_2 = 4$ is excited by the force x_e. The transfer function between the force x_e and the position x_a is given by

$$f(z) = \frac{z^2 - z + 2}{z^4 + 10z^2 + 9}$$



$b^2 - 4(1)(2) = 0$
$r = 5$

Find zeros of function to check where the output signal $x_a(t)$ decays to zero. Also find the singularities of function and characterize whether the singularities are poles or not (Justify your answers by applying definition of pole). [CLO 6]

2. Evaluate $\oint_c \bar{z}\,dz$, where C is a straight line from i to $1 + 2i$. [CLO 4]

3. Evaluate using Cauchy's integral theorem

$$\oint_C \frac{\tan(\frac{z}{2})}{z^4 - 16}\,dz,$$

$(\cos \frac{z}{2})$

where c is the boundary of the square with vertices $\mp 1 \mp i$ (clockwise). [CLO 5]

$z^4 = \sqrt[4]{16}$

$(z^2 - 4)(z^2 + 4)$
$z^2(z^2 + 4) - 4(z^2$
$z^4 - 16$